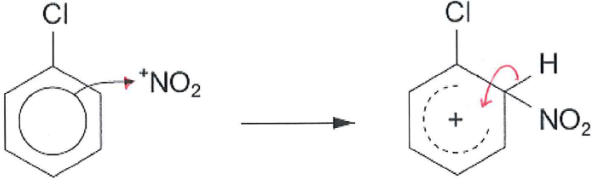


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7.	(a)	(i)		 both curly arrows needed (1) correct structure of intermediate (1)	1	1		2		
		(ii)		electrophilic substitution	1			1		
	(b)			lone pair of electrons on the nitrogen atom(s)	1			1		
	(c)	(i)		orange / red solid	1			1		
		(ii)		the derivatives formed using 2,4-DNPH have suitable melting temperatures for identification / precise melting temperatures			1	1		1
	(d)			dissolve the (impure) oxime in a minimum volume (1) hot methanol / warm methanol (1) water bath / electrical heater (1) allow to cool (1) filter off oxime and dry (1)	1 1	1 1	1	5		5

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(e)			(broad) peak at 3200-3550 cm ⁻¹ due to O—H bond present in the oxime and not in the amide (1) peak at 1650-1750 cm ⁻¹ due C=O bond present in the amide and not in the oxime (1) peak at 3300-3500 cm ⁻¹ due to N—H bond present in the amide and not in the oxime (1)		3		3		
				Question 7 total	6	6	2	14	0	6